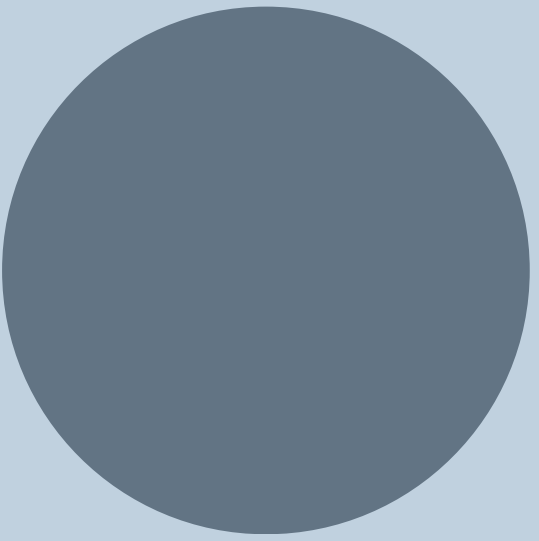




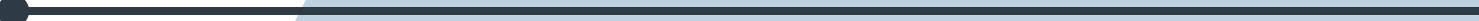
AGILE PROJECT

23BCS014-AKSHETA

23BCS018-ANKITA

23BCS031-DEVADARSHINI

23BCS060-JANANI

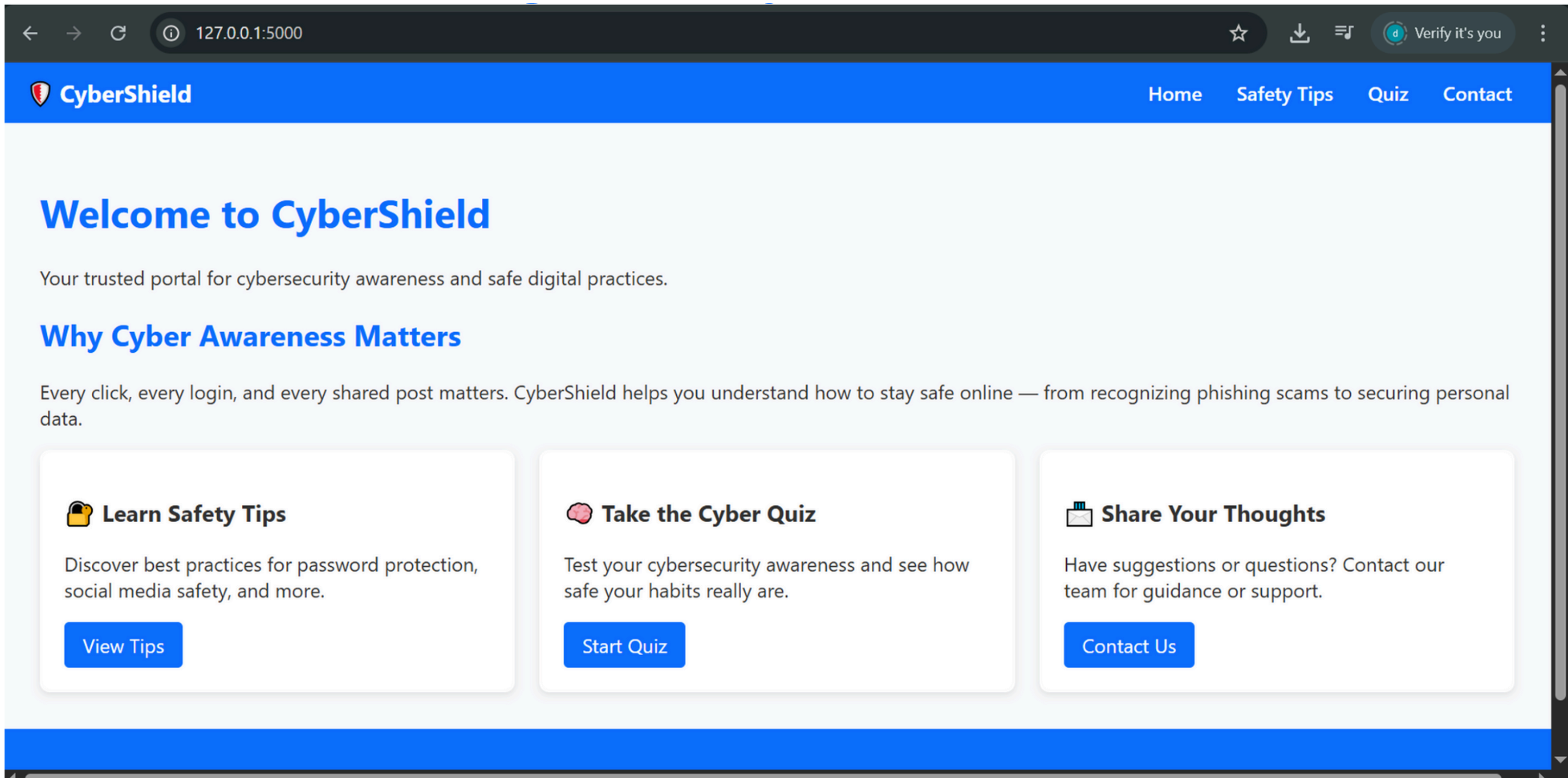


CYBERSHIELD

PROJECT OBJECTIVE :

- To promote awareness about online safety, data privacy, and cyber threats.
- To build a web application that is functional, user-friendly, and secure.
- To thoroughly test the system using different software testing tools and methods.



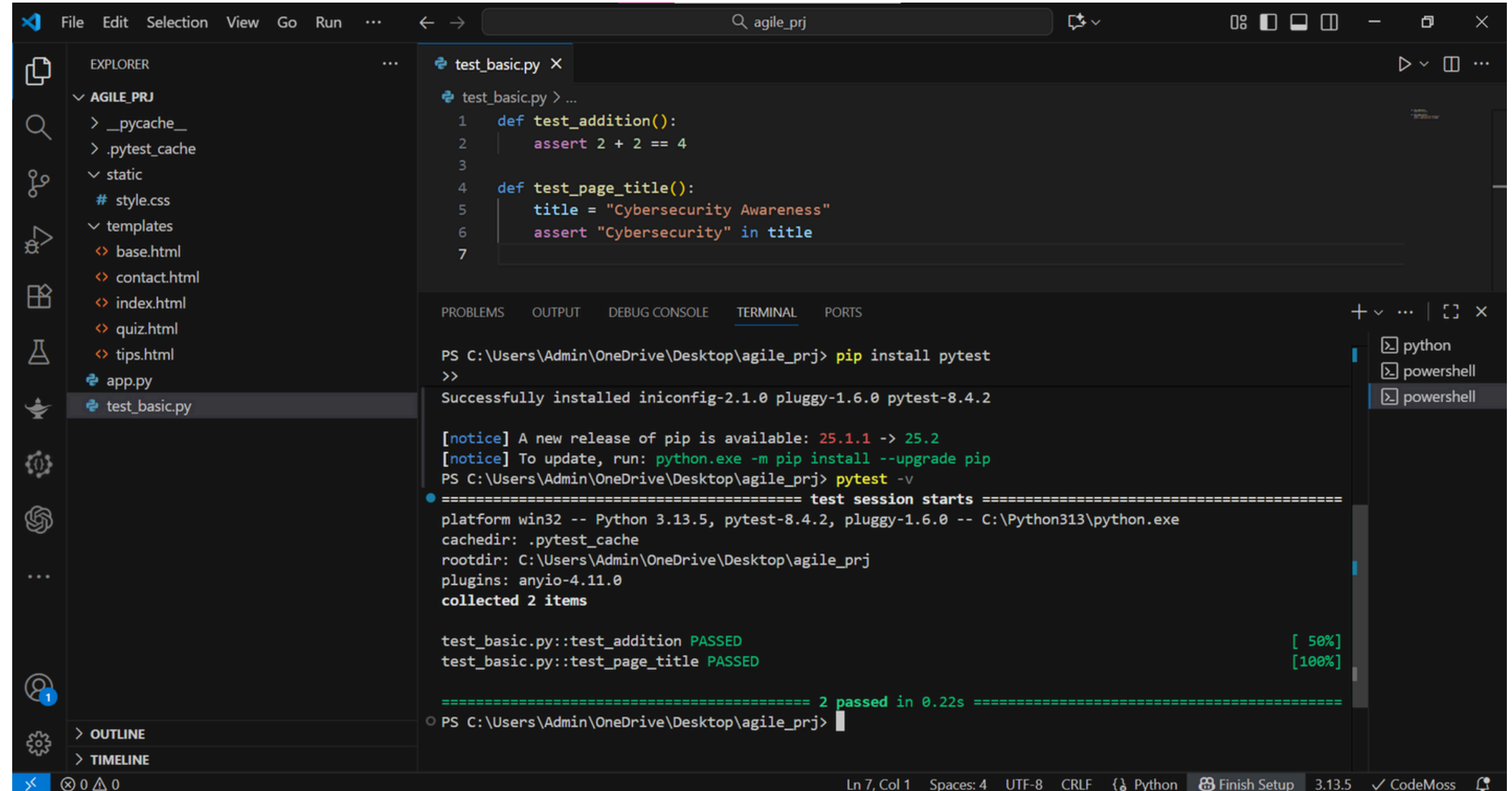


Cybersecurity Safety Tips

- **1. Use Strong Passwords:** Avoid common words; use a mix of letters, numbers, and symbols.
- **2. Beware of Phishing:** Never click suspicious links or attachments in emails.
- **3. Enable Two-Factor Authentication:** Adds extra protection for your accounts.
- **4. Keep Software Updated:** Security patches fix critical vulnerabilities.
- **5. Use Secure Wi-Fi:** Avoid public networks for banking or private activities.

UNIT TESTING – SELENIUM WEBDRIVER (PYTHON)

- Validates individual components or modules of the application.
- Ensures each webpage element like buttons, forms, and links functions properly.
- Helps detect coding errors early in the development cycle.
- Reduces debugging time in later stages.



The screenshot displays the Visual Studio Code interface with a project named 'agile_prj'. The Explorer panel on the left shows the file structure, including 'test_basic.py'. The main editor window shows the code for 'test_basic.py', which contains two test functions: 'test_addition()' and 'test_page_title()'. The terminal panel at the bottom shows the command 'pip install pytest' being executed, followed by 'pytest -v'. The output indicates that both tests passed successfully. The status bar at the bottom shows the current file is 'test_basic.py' at line 7, column 1, with 4 spaces, UTF-8 encoding, and CRLF line endings.

```
test_basic.py
1 def test_addition():
2     assert 2 + 2 == 4
3
4 def test_page_title():
5     title = "Cybersecurity Awareness"
6     assert "Cybersecurity" in title
7
```

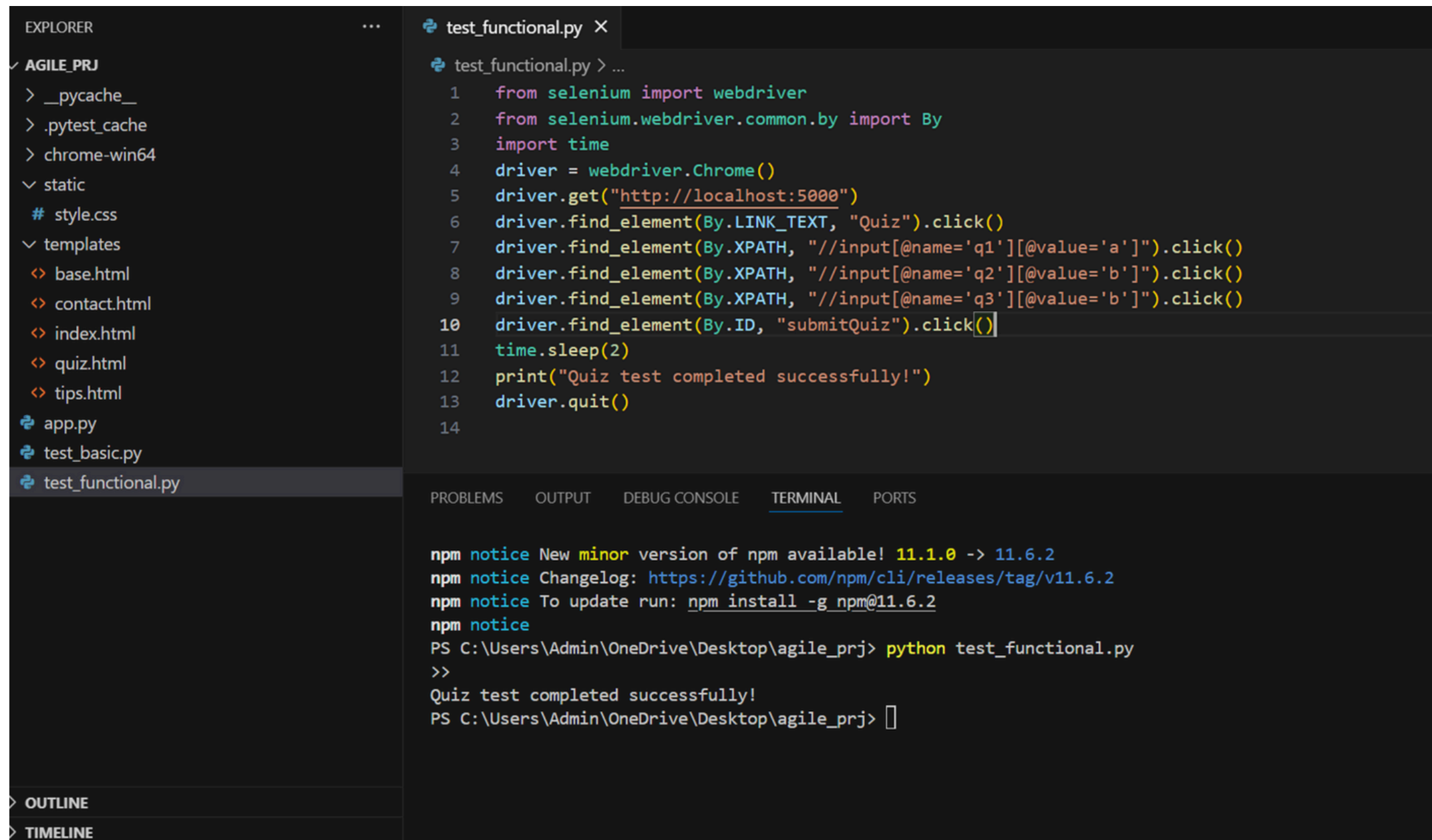
```
PS C:\Users\Admin\OneDrive\Desktop\agile_prj> pip install pytest
>>
Successfully installed iniconfig-2.1.0 pluggy-1.6.0 pytest-8.4.2

[notice] A new release of pip is available: 25.1.1 -> 25.2
[notice] To update, run: python.exe -m pip install --upgrade pip
PS C:\Users\Admin\OneDrive\Desktop\agile_prj> pytest -v
===== test session starts =====
platform win32 -- Python 3.13.5, pytest-8.4.2, pluggy-1.6.0 -- C:\Python313\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\Admin\OneDrive\Desktop\agile_prj
plugins: anyio-4.11.0
collected 2 items

test_basic.py::test_addition PASSED [ 50%]
test_basic.py::test_page_title PASSED [100%]

===== 2 passed in 0.22s =====
PS C:\Users\Admin\OneDrive\Desktop\agile_prj>
```

FUNCTIONAL TESTING – PYTEST (PYTHON FRAMEWORK)



The screenshot displays a Visual Studio Code interface. On the left, the Explorer pane shows a project named 'AGILE_PRJ' with files like 'style.css', 'base.html', 'contact.html', 'index.html', 'quiz.html', 'tips.html', 'app.py', 'test_basic.py', and 'test_functional.py'. The main editor area shows the content of 'test_functional.py', which is a Selenium WebDriver script. The script imports 'webdriver' from 'selenium', 'By' from 'selenium.webdriver.common.by', and 'time'. It initializes a Chrome driver, navigates to 'http://localhost:5000', clicks a 'Quiz' link, fills out a form with values 'a', 'b', and 'b' for three input fields, clicks a 'submitQuiz' button, sleeps for 2 seconds, prints 'Quiz test completed successfully!', and finally quits the driver. Below the editor, the TERMINAL pane shows the command 'python test_functional.py' being executed, resulting in the output 'Quiz test completed successfully!'.

```
test_functional.py
1 from selenium import webdriver
2 from selenium.webdriver.common.by import By
3 import time
4 driver = webdriver.Chrome()
5 driver.get("http://localhost:5000")
6 driver.find_element(By.LINK_TEXT, "Quiz").click()
7 driver.find_element(By.XPATH, "//input[@name='q1'][@value='a']").click()
8 driver.find_element(By.XPATH, "//input[@name='q2'][@value='b']").click()
9 driver.find_element(By.XPATH, "//input[@name='q3'][@value='b']").click()
10 driver.find_element(By.ID, "submitQuiz").click()
11 time.sleep(2)
12 print("Quiz test completed successfully!")
13 driver.quit()
14
```

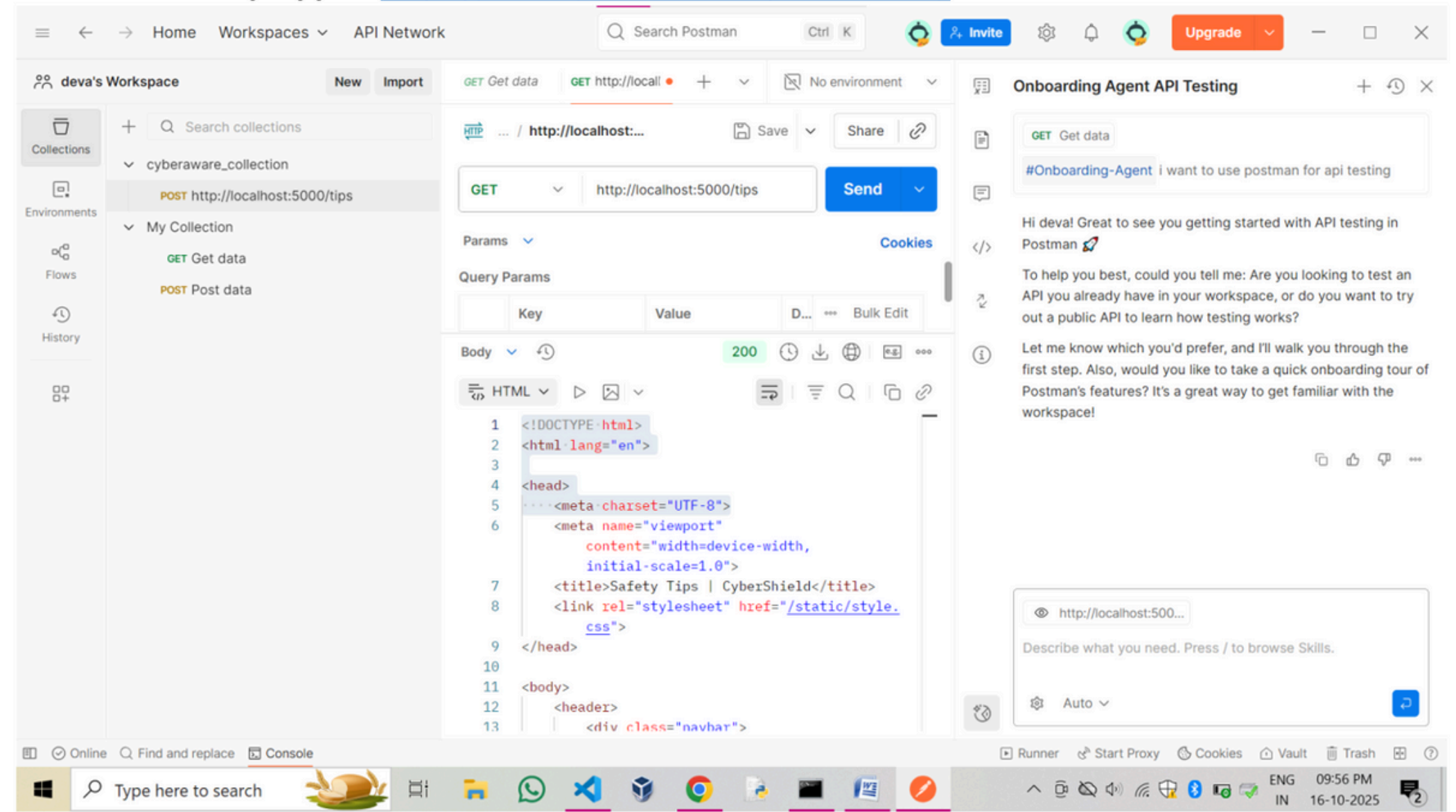
```
npm notice New minor version of npm available! 11.1.0 -> 11.6.2
npm notice Changelog: https://github.com/npm/cli/releases/tag/v11.6.2
npm notice To update run: npm install -g npm@11.6.2
npm notice
PS C:\Users\Admin\OneDrive\Desktop\agile_prj> python test_functional.py
>>
Quiz test completed successfully!
PS C:\Users\Admin\OneDrive\Desktop\agile_prj>
```

- Checks if the application performs according to user requirements.
- Validates login, registration, and navigation processes.
- Confirms correct input-output behavior for each function.
- Ensures overall system functionality and user flow are error-free.

API TESTING – POSTMAN

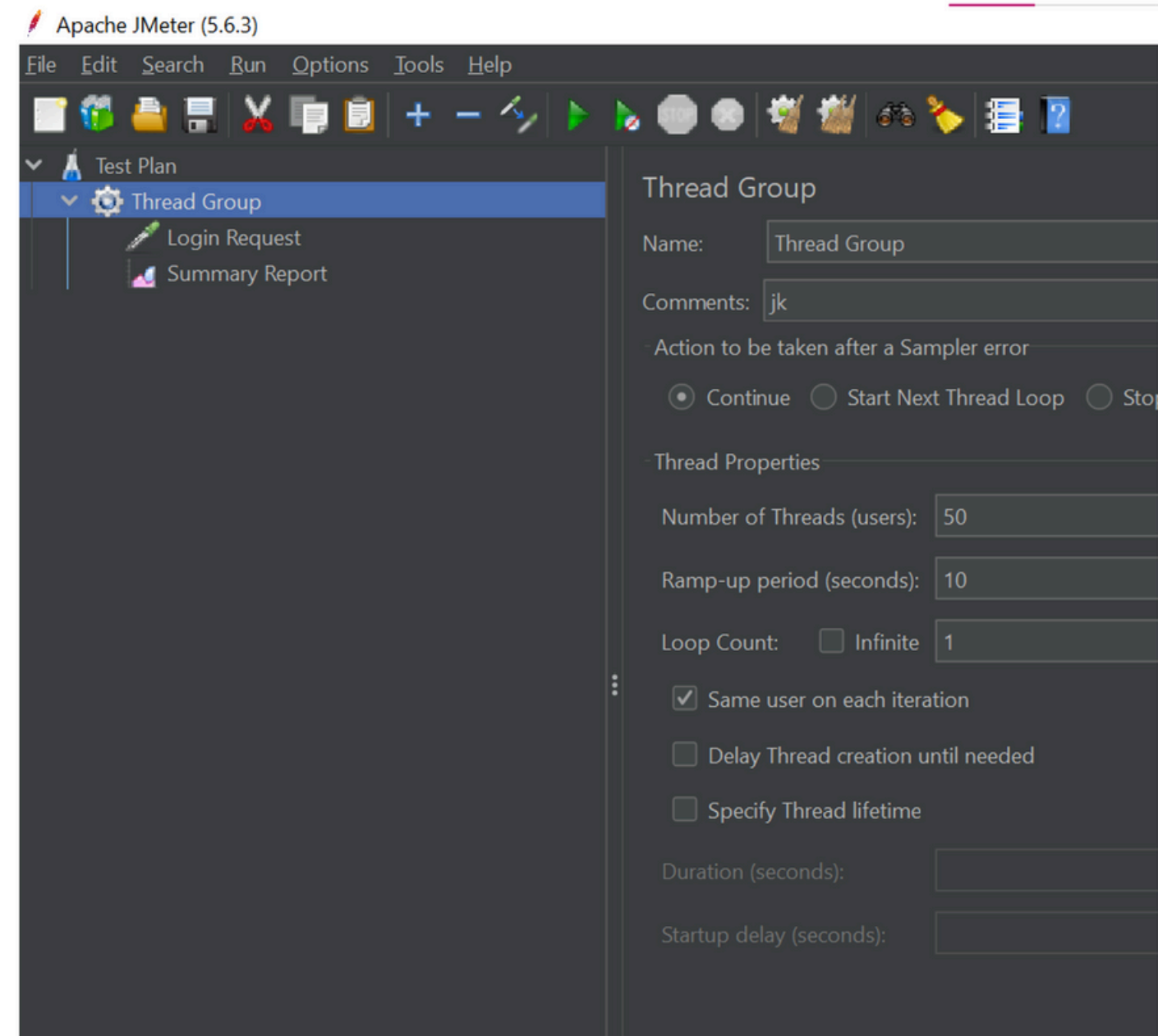
- Verifies communication between frontend and backend systems.
- Tests all endpoints to ensure accurate data transfer and responses.
- Checks API performance, response time, and error handling.
- Ensures consistent and secure exchange of information.

Postman Desktop App → <https://www.postman.com/downloads/>



PERFORMANCE TESTING – APACHE JMETER

- Download → https://jmeter.apache.org/download_jmeter.cgi



- Tests how the application performs under heavy load conditions.
- Simulates multiple users accessing the app simultaneously.
- Measures response time, speed, and resource usage.
- Identifies bottlenecks affecting system performance.

Summary Report

Name: Summary Report

Comments:

Write results to file / Read from file

Filename

Browse...

Log/Display Only: ☐ Errors ☐ Successes

Configure

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received KB...	Sent KB/sec	Avg. Bytes
Login Request	159	9	3	147	14.13	0.00%	62928624454	0.45	0.29	370.0
TOTAL	159	9	3	147	14.13	0.00%	1.3/sec	0.45	0.29	370.0

Test Plan

Thread Group

Login Request

Summary Report

HTTP Request

Name: Login Request

Comments:

BasicAdvanced

Web Server

Protocol [http]: httpServer Name or IP: localhost

HTTP Request

POSTPath: /tips

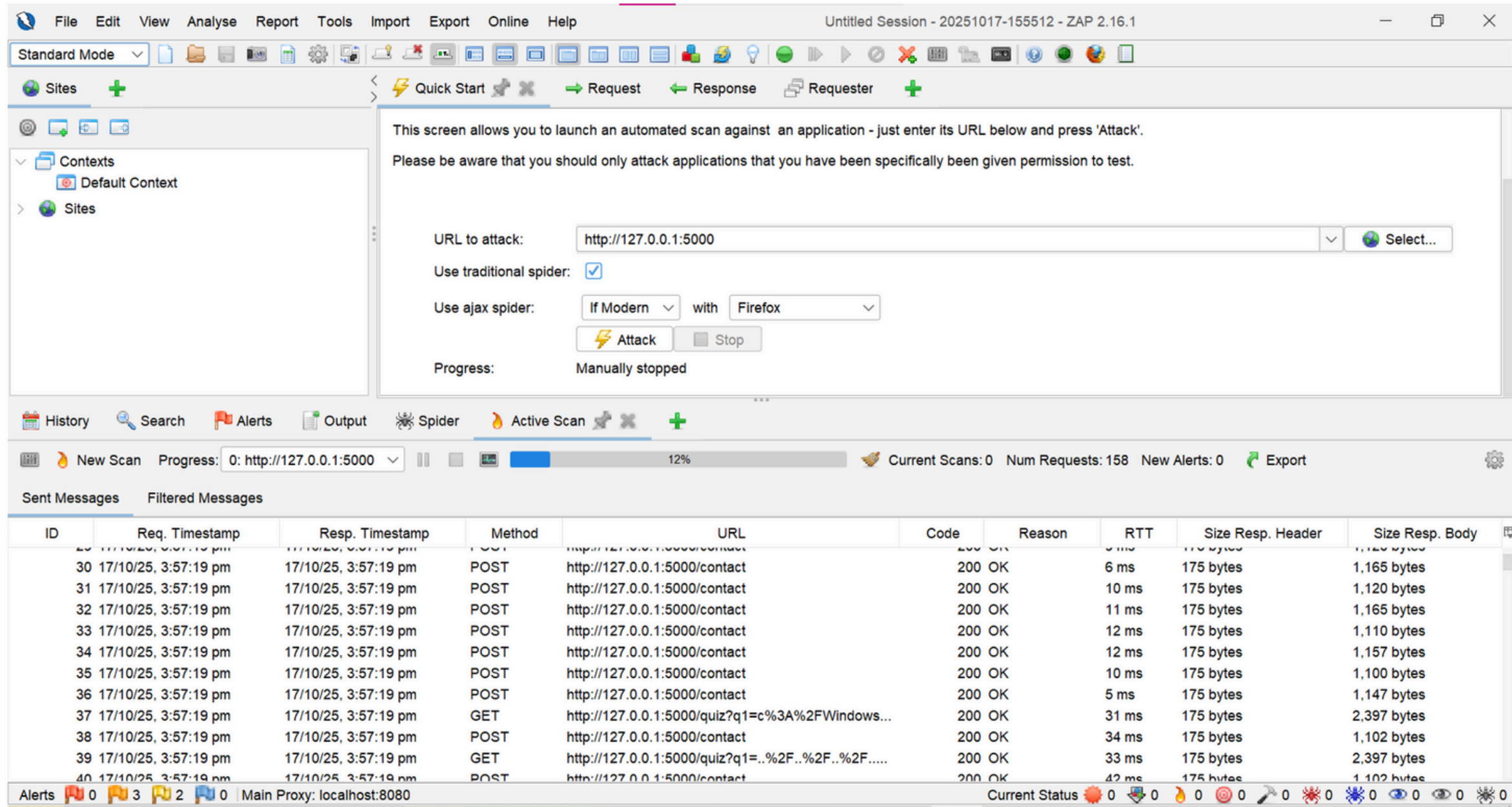
☐ Redirect Automatically☒ Follow Redirects☒ Use KeepAlive☐ Use mu

ParametersBody DataFiles Upload

Name:	Value
username	admin
password	admin

Send

SECURITY TESTING – OWASP ZAP



The screenshot displays the OWASP ZAP 2.16.1 interface. The top menu bar includes File, Edit, View, Analyse, Report, Tools, Import, Export, Online, and Help. The main window is titled "Untitled Session - 20251017-155512 - ZAP 2.16.1". The left sidebar shows a tree view with "Contexts" (Default Context) and "Sites". The main area is the "Quick Start" screen, which provides instructions on how to launch an automated scan. It includes a "URL to attack:" field with the value "http://127.0.0.1:5000", a "Use traditional spider:" checkbox (checked), and a "Use ajax spider:" dropdown menu (set to "If Modern" with "Firefox" selected). There are "Attack" and "Stop" buttons, and a "Progress:" indicator showing "Manually stopped".

Below the Quick Start screen, there is a "History" tab showing a list of HTTP requests. The table has columns for ID, Req. Timestamp, Resp. Timestamp, Method, URL, Code, Reason, RTT, Size Resp. Header, and Size Resp. Body. The table contains 15 rows of data, all with a status of "200 OK".

ID	Req. Timestamp	Resp. Timestamp	Method	URL	Code	Reason	RTT	Size Resp. Header	Size Resp. Body
29	17/10/25, 3:57:19 pm	17/10/25, 3:57:19 pm	POST	http://127.0.0.1:5000/contact	200	OK	6 ms	175 bytes	1,120 bytes
30	17/10/25, 3:57:19 pm	17/10/25, 3:57:19 pm	POST	http://127.0.0.1:5000/contact	200	OK	6 ms	175 bytes	1,165 bytes
31	17/10/25, 3:57:19 pm	17/10/25, 3:57:19 pm	POST	http://127.0.0.1:5000/contact	200	OK	10 ms	175 bytes	1,120 bytes
32	17/10/25, 3:57:19 pm	17/10/25, 3:57:19 pm	POST	http://127.0.0.1:5000/contact	200	OK	11 ms	175 bytes	1,165 bytes
33	17/10/25, 3:57:19 pm	17/10/25, 3:57:19 pm	POST	http://127.0.0.1:5000/contact	200	OK	12 ms	175 bytes	1,110 bytes
34	17/10/25, 3:57:19 pm	17/10/25, 3:57:19 pm	POST	http://127.0.0.1:5000/contact	200	OK	12 ms	175 bytes	1,157 bytes
35	17/10/25, 3:57:19 pm	17/10/25, 3:57:19 pm	POST	http://127.0.0.1:5000/contact	200	OK	10 ms	175 bytes	1,100 bytes
36	17/10/25, 3:57:19 pm	17/10/25, 3:57:19 pm	POST	http://127.0.0.1:5000/contact	200	OK	5 ms	175 bytes	1,147 bytes
37	17/10/25, 3:57:19 pm	17/10/25, 3:57:19 pm	GET	http://127.0.0.1:5000/quiz?q1=c%3A%2FWindows...	200	OK	31 ms	175 bytes	2,397 bytes
38	17/10/25, 3:57:19 pm	17/10/25, 3:57:19 pm	POST	http://127.0.0.1:5000/contact	200	OK	34 ms	175 bytes	1,102 bytes
39	17/10/25, 3:57:19 pm	17/10/25, 3:57:19 pm	GET	http://127.0.0.1:5000/quiz?q1=..%2F..%2F..%2F.....	200	OK	33 ms	175 bytes	2,397 bytes
40	17/10/25, 3:57:19 pm	17/10/25, 3:57:19 pm	POST	http://127.0.0.1:5000/contact	200	OK	42 ms	175 bytes	1,102 bytes

At the bottom, there is an "Alerts" tab showing 0 alerts. The status bar at the very bottom indicates "Current Status" with various icons and counts: 0 alerts, 3 warnings, 2 errors, 0 info, 0 main proxy, 0 local host, 0 current status, 0 fire, 0 bomb, 0 spider, 0 ajax, 0 http, 0 https, 0 ssl, 0 tls, 0 dns, 0 smtp, 0 imap, 0 ftp, 0 ssh, 0 sftp, 0 scp, 0 svn, 0 git, 0 svn, 0 git, 0 svn, 0 git.

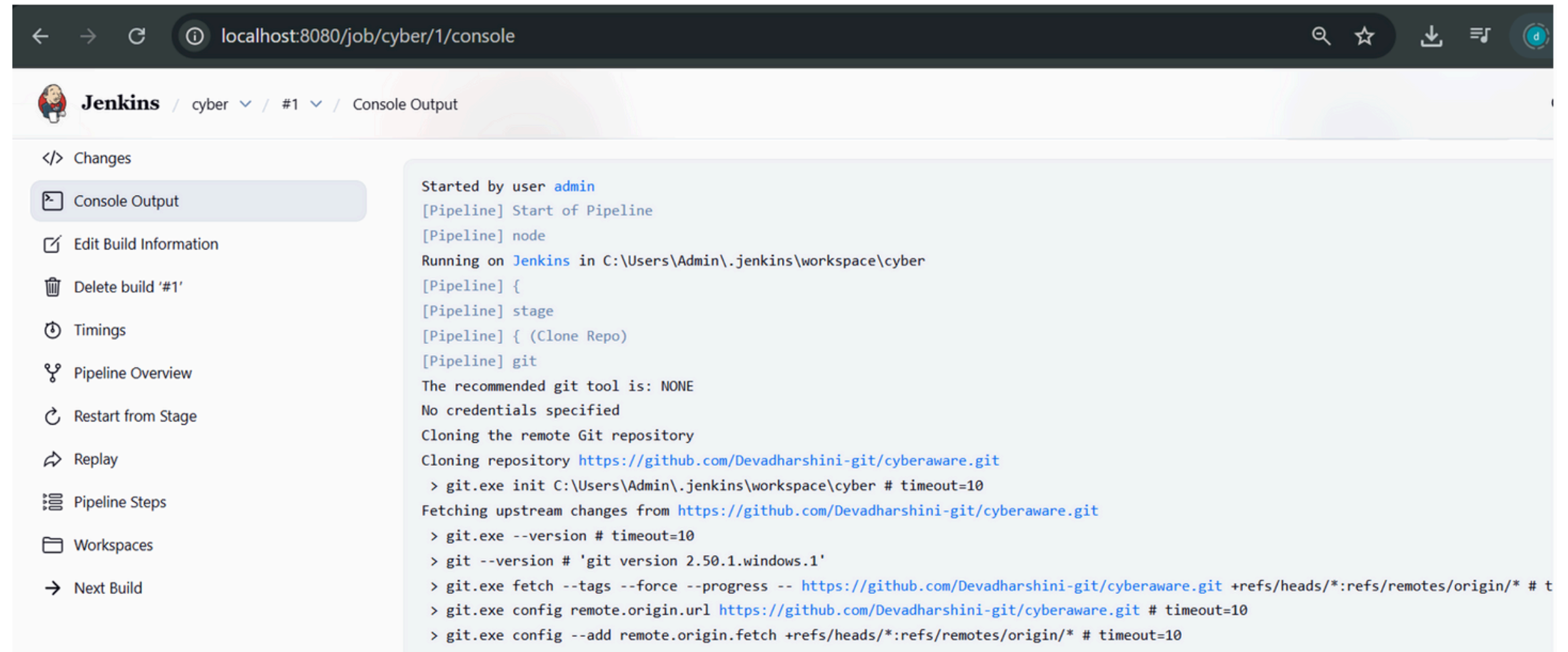
- Scans for vulnerabilities like SQL Injection, XSS, and insecure authentication.
- Evaluates data protection mechanisms within the application.
- Ensures secure communication and prevents data breaches.

	A	B	C	D	E	F	G	H	I	J	K
	ID	Req. Time	Resp. Time	Method	URL	Code	Reason	RTT	Size Resp.	Size Resp.	Body
	24	Fri Oct 17	Fri Oct 17	GET	http://127	404	NOT FOUND	5	181	207	
	26	Fri Oct 17	Fri Oct 17	GET	http://127	404	NOT FOUND	5	181	207	
	28	Fri Oct 17	Fri Oct 17	GET	http://127	404	NOT FOUND	6	181	207	
	29	Fri Oct 17	Fri Oct 17	POST	http://127	200	OK	9	175	1120	
	30	Fri Oct 17	Fri Oct 17	POST	http://127	200	OK	6	175	1165	
	31	Fri Oct 17	Fri Oct 17	POST	http://127	200	OK	10	175	1120	
	32	Fri Oct 17	Fri Oct 17	POST	http://127	200	OK	11	175	1165	
	33	Fri Oct 17	Fri Oct 17	POST	http://127	200	OK	12	175	1110	
	34	Fri Oct 17	Fri Oct 17	POST	http://127	200	OK	12	175	1157	
	35	Fri Oct 17	Fri Oct 17	POST	http://127	200	OK	10	175	1100	
	36	Fri Oct 17	Fri Oct 17	POST	http://127	200	OK	5	175	1147	
	37	Fri Oct 17	Fri Oct 17	GET	http://127	200	OK	31	175	2397	
	38	Fri Oct 17	Fri Oct 17	POST	http://127	200	OK	34	175	1102	
	39	Fri Oct 17	Fri Oct 17	GET	http://127	200	OK	33	175	2397	
	40	Fri Oct 17	Fri Oct 17	POST	http://127	200	OK	42	175	1102	
	41	Fri Oct 17	Fri Oct 17	GET	http://127	200	OK	46	175	2397	
	42	Fri Oct 17	Fri Oct 17	POST	http://127	200	OK	43	175	1114	
	43	Fri Oct 17	Fri Oct 17	GET	http://127	200	OK	36	175	2397	
	44	Fri Oct 17	Fri Oct 17	POST	http://127	200	OK	35	175	1114	
	45	Fri Oct 17	Fri Oct 17	GET	http://127	200	OK	37	175	2397	
	46	Fri Oct 17	Fri Oct 17	POST	http://127	200	OK	20	175	1115	
	47	Fri Oct 17	Fri Oct 17	GET	http://127	200	OK	31	175	2397	
	48	Fri Oct 17	Fri Oct 17	POST	http://127	200	OK	40	175	1115	

- Strengthens overall application security before deployment.
- Download → <https://www.zaproxy.org/download/>

CONTINUOUS INTEGRATION TESTING – JENKINS

- Automates the testing and deployment process for every code update.
- Detects build issues early and notifies the team automatically.
- Maintains code consistency and stability across versions.
- Supports Agile principles of continuous delivery and improvement.
- Download → <https://www.jenkins.io/download/>



The screenshot shows the Jenkins web interface in a browser. The address bar indicates the URL is `localhost:8080/job/cyber/1/console`. The page title is "Jenkins / cyber / #1 / Console Output". On the left sidebar, the "Console Output" tab is selected. The main area displays the following log output:

```
Started by user admin
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\Users\Admin\.jenkins\workspace\cyber
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Clone Repo)
[Pipeline] git
The recommended git tool is: NONE
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/Devadharshini-git/cyberaware.git
> git.exe init C:\Users\Admin\.jenkins\workspace\cyber # timeout=10
Fetching upstream changes from https://github.com/Devadharshini-git/cyberaware.git
> git.exe --version # timeout=10
> git --version # 'git version 2.50.1.windows.1'
> git.exe fetch --tags --force --progress -- https://github.com/Devadharshini-git/cyberaware.git +refs/heads/*:refs/remotes/origin/* # t
> git.exe config remote.origin.url https://github.com/Devadharshini-git/cyberaware.git # timeout=10
> git.exe config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
```

RESULT AND INFERENCE

- All test cases passed successfully, confirming overall system stability.
 - Functional and Unit Testing verified that all pages and features worked as expected.
 - API Testing confirmed accurate responses and proper communication between frontend and backend.
 - Performance Testing (JMeter) showed the system can efficiently handle multiple simultaneous users.
 - Security Testing (OWASP ZAP) detected no major or critical vulnerabilities.
 - Mobile Testing ensured the web application is responsive and works smoothly across all devices.
 - GUI Testing validated all buttons, forms, and interactive elements function correctly.
 - Behavior-Driven Testing confirmed that user workflows performed as intended.
 - Overall, the application is stable, secure, fully functional, and ready for deployment and demonstration.
-

CONCLUSION

- The Agile testing approach ensured continuous improvement and early detection of issues.
- All testing stages — unit, functional, API, performance, and security — were successfully completed.
- The application proved to be stable, secure, and efficient under different test conditions.
- No critical bugs or vulnerabilities were found during the testing process.
- The project achieved its main goal of building a Cybersecurity Awareness Web Application that is reliable and user-friendly.
- The system is ready for deployment and real-world demonstration, ensuring a strong foundation for promoting cyber awareness.





**Thank
You**

